



RawBeam 350 marks the next step in Ayrton's laser-source development, following over 15 years of work with laser phosphor technology. It has been designed as a practical replacement to the gradual disappearance of traditional discharge-lamp sources in beam fixtures. Rather than simply recreating the past, RawBeam 350 uses a modern source designed to deliver the familiar energy and precision expected from this type of light.

The fixture brings together our experience in optics, thermal management, colour systems and compact mechanical design. Its feature set has been developed for real-world use, with the tools needed for beam work, projection, aerial effects and touring applications. RawBeam 350 is therefore not just a replacement for an older technology, but a carefully developed platform for the next generation of beam lights.

★ KEY FEATURES

➤ Small

RawBeam 350 is a powerful Phosphor laser-source beam fixture in an extremely compact format. It is as small as Diablo, which has long been considered a reference point for compact moving lights.

➤ Light

At 25.4 kg, RawBeam 350 is lighter and easier to handle than comparable products. This makes rigging, transport and daily use simpler, especially when deployed in large quantities.

➤ Quiet

RawBeam 350 has been designed with highly efficient thermal management. It can cool the engine effectively without needing to run the fans at high speed, making it noticeably quieter in real show conditions.

➤ Narrow

RawBeam 350 reaches an extremely tight 1.1° beam in Beam Mode. This gives sharp, concentrated aerial effects with strong intensity and definition.

➤ Wide

RawBeam 350 opens up to 45° in Spot Mode. This gives more flexibility for wider shots, gobos, prisms and projected effects.

➤ Bright

With its 350 W laser phosphor source, RawBeam 350 delivers very high intensity from a compact fixture. It gives the beam the punch and energy expected from a discharge-style source, with the efficiency and control of laser phosphor technology.

TECHNICAL SPECIFICATIONS

SOURCE	
Engine	LASER Phosphore
LASER Watts	350 W (total fixture use, see Electrical section)
LASER Engine Life (L70)	Up to 12,000 hours
LASER Refresh Rate	1.2 kHz, 2.4 kHz, 16 kHz
Dimming	Electronic
Dimming Curves	Square & Linear
Dimming Control Options	Auto on/off dimmer smoothing when Snap effects
Strobe Speed	1-25 Hz Per Second
Strobe Modes	Strobe Effects / Pulse Effects / Random Effects

ELECTRICAL	
Max Power Consumption	660 W
Standby Consumption	94.5 W @ 20°C
Voltage Range	100-240 V 50-60 Hz
Current (Max)	6 A @ 230 V / 2.9 A @ 110 V
Inrush (Half Cycle)	< 20 A @ 230 V
Maximum Current available at the output	16 A
Connector	REAN Power X - RRAC3M In/Thru
Power Supply Unit	Active PFC
Contains Battery	N/A

TYPICAL POWER AND CURRENT	
120V , 60Hz	565W, 4.753A, PF 0.989
208V, 60Hz	550W , 2.714A , PF 0.973
230V, 50Hz	549W , 2.475A , PF 0.964
240V, 50Hz	548W , 2.382A , PF 0.958

Suitable disclaimer nominal, etc

OPTICAL PERFORMANCE	
Lens Size	160 mm (6.29 in)
Zoom Range	Beam Mode: 1.1° to 18° Spot Mode: 2.2° to 45°
Zoom Ratio	40:1
Gate Size	4 mm

MOVEMENT		
Pan Range	Infinite	
Pan Speed (End to End or range)	180°	322.3 rpm
	360°	322.3 rpm
	EtoE	322.3 rpm
Tilt Range	Infinite	
Tilt Speed (End to End or range)	180°	281.3 rpm
	360°	281.3 rpm
	EtoE	281.3 rpm
Resolution	8- or 16-bit	
Calibration Time	55.6 sec	
Position Correction System	Yes / Automatic repositioning	
Follow Spot Modes	Manual & Tracking system	

COLOUR SYSTEM	
Colour System(s)	CMY + CTO
Control	CMY + CTO 16-bit
CTO	3000 K to 6800 K
Variable CTP	N/A
Colour Wheel	15 complementary colours / 1 multicolour section / 1 CTB
Colour Wheel Effects	Continuous Rotation

GOBOS	
Indexing Wheel	1
Indexing Gobos	10 (quartz, see load below) + open position
Gobo Diameter	16 mm (0.63 in)
Image Diameter	10 mm (0.40 in)
Static Wheel	1
Static Gobos	25 metal gobo + 4 beam reducers + open
Animation Wheel	Yes (on the Static gobo wheel)
Animation Details	Bi-directional with continuous rotation



TECHNICAL SPECIFICATIONS

PERFORMANCE		
LASER engine output (Sphere)		22,000 lm
Fixture Output (Sphere)		18,000 lm (Fan Mode Auto)
		18,200 lm (Fan Mode Stage)
		13,200 lm (Fan Mode Silence)
		11,300 lm (Fan Mode Super Silence)
Fixture Output (Goniometer)		18,100 lm (Fan Mode Auto, after derating)
Field Lumens (Goniometer)	Narrow	13,700 lm
	Middle	16,000 lm
	Wide	18,000 lm
Lumen Per Watt (LPW) Max fixture output / 230 V Typical power		TBA
CCT		6,800 K
Δuv		0.013
CRI		Ra: 66 (+/-2)
With TrueColour™ Filter		N/A
Tm30		64,4
TLCI		38
CQS		61,4

IMAGING	
Framing	N/A
Framing Rotation	N/A
Iris	None (4 Beam reducers on the fixed gobo wheel)
Frost	2 Frost : light 1° & heavy 5°
Prism	2 combinable sets of 3 rotating and indexable prisms

CONNECTORS	
Power	REAN Power X - RRAC3M In/Thru
DMX	IP65 In/Thru 5 Pin XLR
Network	IP65 In/Thru RJ45
Wireless	DMX Only

CONTROL	
DMX Control	DMX (USITT DMX-512-A) RDM (ANSI/ESTA E1.20)
Transceiver/Receiver	RS-485 with Opto Isolation
DMX Modes	1 User Mode
Channel Count	Extended - 43
DMX Chart	Scan the QR code on page 5
Network Protocols	ArtNet & sACN
Local Control	Colour LCD Panel 5 Menu Buttons NFC
Stand Alone Mode	Leader/Follower, presets 250
RDM Functions	Yes
Wireless DMX	CRMx TiMo RDM receiver
E-Sim Ready	No
USB Port	No

THERMAL	
Max Surface Temperature, steady state, at Ta 40°	85 °C (185 °F)
Max Ambient Temperature (Ta Max)	45 °C (113 °F)
Min Ambient Temperature (Ta Min)	-20 °C (-4 °F)
Total Heat Dissipation (Calculated +/- 10%)	1870 BTU/H @ 240V
	1930 BTU/H @ 120V
Minimum to illuminated surface	20 m (65.6 ft)
Minimum to combustible surface	0.1 m (0.32 ft)
Thermal Protection system	Yes
Cooling modes	See Acoustics
Cooling System	Advanced Liquid Cooling IP68 Self adjusting Variable speed fans

TECHNICAL SPECIFICATIONS

ACOUSTICS STATIC

Fan Mode	dBA ¹	NC ²
Auto	TBA	TBA
Stage	TBA	TBA
Silence/Studio	TBA	TBA
Super Silence	TBA	TBA

DYNAMIC

Mode	dBA ¹	NC ²
Auto	TBA	TBA
Stage	TBA	TBA
Silence/Studio	TBA	TBA
Super Silence	TBA	TBA

1. Measured @ 1m Static, 31dBA ambient

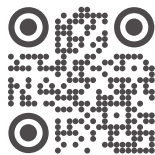
2. Noise Criterion

https://www.engineeringtoolbox.com/nc-noise-criterion-d_517.html

ENVIRONMENTAL

IP Rating	IP 65
Salt Spray Testing	Pending
Estimated embodied carbon including: ▪ A1 – Raw materials of fixture and packaging ▪ A3 – Product manufacturing	296.7 kg CO ₂ e/fixture
Recycling options	Packaging widely recyclable. Cardboard is made of recycled pulp ³ .
SIP Optional	Yes

Our Environmental Commitment



3. Up to 92%

WARRANTY

Fixture	2 years
LASER Engine	2 years

From date of invoice

INSTALLATION

Orientation	Any
Mounting attachments	- Fastening bracket system: two Omega ¼ turn brackets designed for use with standard clamps - Mounting points: four ¼ turn locking fittings allowing installation of Omega brackets on luminaire - Safety cable attachment point
Installation version	Yes, Fixed Tails

CONSTRUCTION

Colour	Carbon Black Standard Custom colours on request
Materials	- Moving-head skeleton made of aluminium and steel metal plates - Base in die-cast aluminium - Heatsinks in aluminium and copper - Moulded covers in self-extinguishing fire retardant ABS PC (VO class) - Two-side handles for transportation - Four heavy-duty feet for better stability

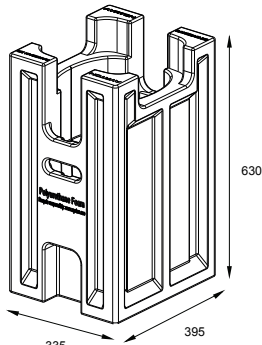
INCLUDED ITEMS

Manual	1
OMEGA brackets with quarter-turn locks	2
Cables	- IP65: 1 cable PowerCON True1 TOP to bare end, 1.5m - IP66: DMX IN / OUT tails to bare end for IP66 DMX connection - Ethernet IN / OUT tails to bare end for IP66 ArtNet™ connection - Power tail to bare end for IP66 power connection
Other accessories	N/A

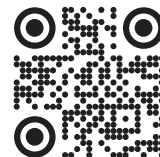


TECHNICAL SPECIFICATIONS

PHYSICAL

Fixture / IP65 / IP66	
Length	339 mm (13.35 in)
Height	592 mm (23.31 in)
Depth	297 mm (11.70 in)
Weight	25.4 kg (55.11 lb)
Shipping Dimensions	
Length	405 mm (15.95 in)
Height	705 mm (27.75 in)
Depth	340 mm (13.38 in)
Weight	36.4 kg (80.25 lb)
In SIP Dimensions	
Length	395 mm (15.55 in)
Height	630 mm (24.80 in)
Depth	335 mm (13.20 in)
Weight (excl fixture)	11 kg (24.25 lb)
SIP Drawing	

SUPPORTING DOCUMENTS

[Manual](#)[GDTF Profile](#)[GMA Profile \(TBC\)](#)[IES FILE \(TBC\)](#)[DMX Tables](#)[Photometric Files](#)[Software](#)[Warranty](#)

COMPLIANCE

EU (EMC & LVD)

- EMC Directive 2014/30/EU
- LVD Directive 2014/35/EU

US (Safety & EMC)

- UL 1573
- FCC Part 15, Subpart B: 2021

Substance Regulations

- RoHS
- REACH
- CP65

Declarations

Scan the QR code above

ORDERING DETAILS

RawBeam 350

014350

ACCESORIES

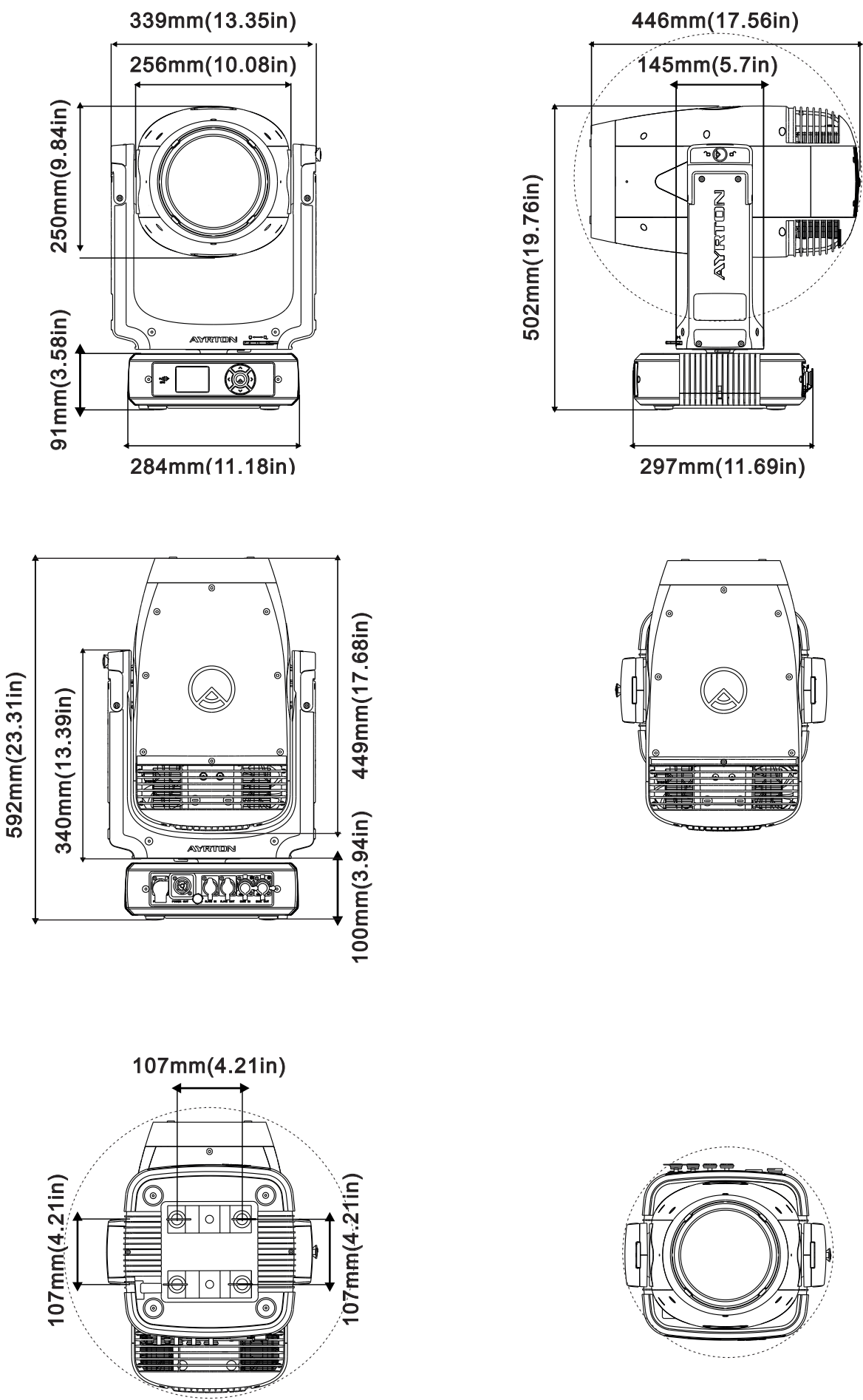
IP Test Kit

048250

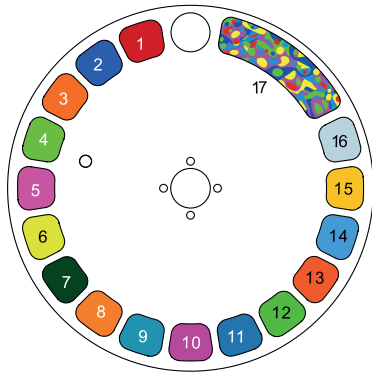
Lynx Control

045130

TECHNICAL SPECIFICATIONS

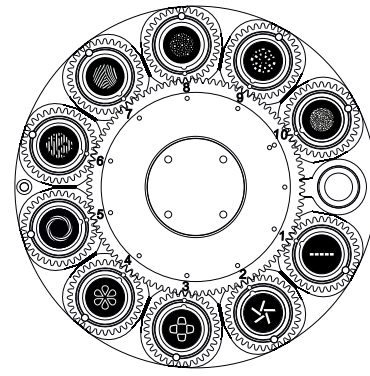


TECHNICAL SPECIFICATIONS



COLOUR WHEEL

1	Red
2	Blue
3	Orange
4	Green
5	Pink
6	Yellow
7	Velvet Green
8	Amber
9	Light Blue
10	Follies Pink
11	Slate Blue
12	Dark Green
13	Dark Amber
14	Medium Blue
15	Oklahoma Yellow
16	CTV
17	Colour animation



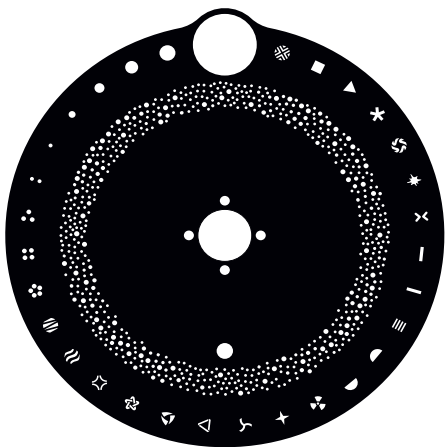
GOBO WHEEL 1

Rotating Gobo

1	364	Rectangle Line 5	GPG0501727
2	344	Star Bar 5	GPG0501725
3	186	Game Pad Stroke	GPG0501730
4	264	Daisy Stroke	GPG0501723
5	112	Nested Rings	GPG0501736
6	278	Bread Stix	GPG0501751
7	096	Zebra	GPG0501752
8	075	Starfield	GPG0501732
9	070	Circle of Squares	GPG0501724
10	101	Deep Forest	GPG0501728

This side facing front lens

TECHNICAL SPECIFICATIONS



GOBO WHEEL 2

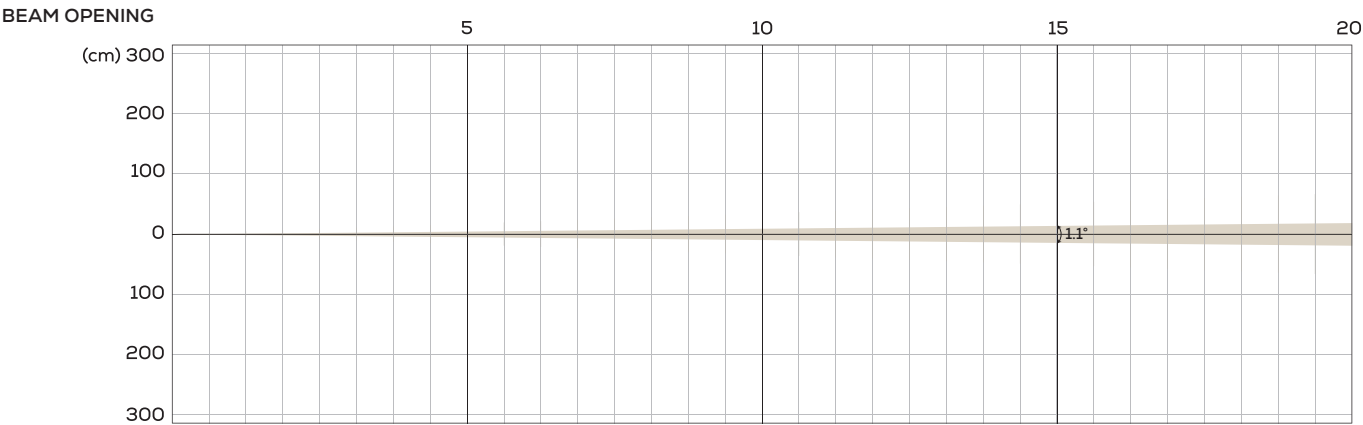
Static Gobo

1	302	80% Iris Beam	9	257	Zig Zag Light	17	149	Nuclear	25	439	Circular Saw 4
2	304	60% Iris Beam	10	099	Waves Light	18	299	Half Beam Right	26	394	Starfield Light
3	306	40% Iris Beam	11	419	Iron Ball Light	19	298	Half Beam Left	27	340	Triangle Beam
4	308	20% Iris Beam	12	373	Nested Star	20	353	Four Lines H	28	330	Square Beam
5	312	Dot Line 2	13	261	Arrow Ring 3	21	350	Bold Line	29	435	Nested Cross
6	326	Dot Triangle 3	14	427	Split Triangle 2	22	351	Vertical Line			
7	328	Dot Square 4	15	117	Helix 3	23	437	Split Cross			
8	319	Dot Ring 5	16	374	Compass 4	24	270	Ink Stain 2			

This side facing front lens

AUTO MODE: PHOTOMETRIC DETAIL

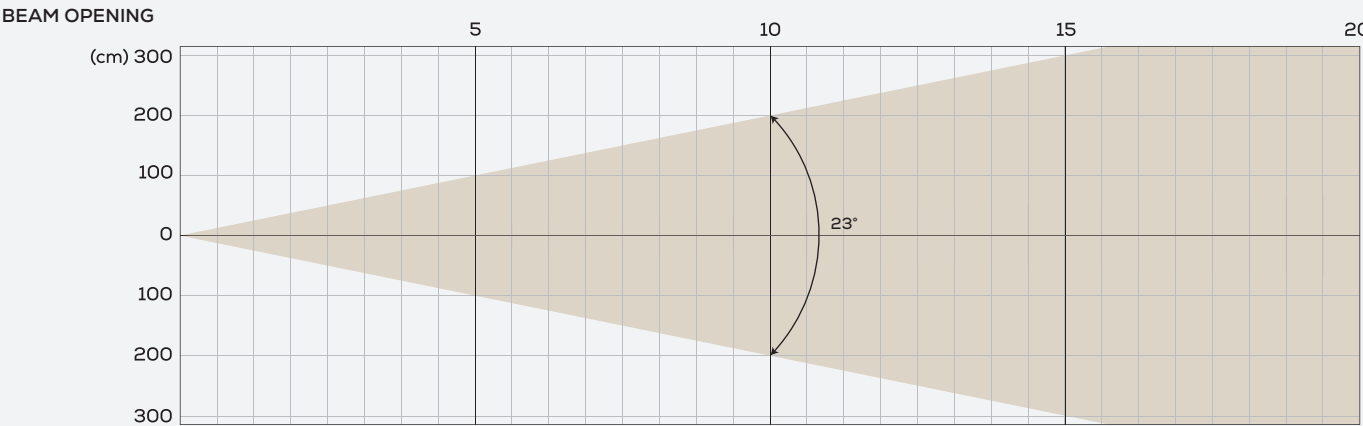
MIN ZOOM



INTENSITY

Distance (m)	5		10		15		20		30		50		100	
unit	Lux	Diameter	Lux	Diameter	Lux	Diameter	Lux	Diameter	Lux	Diameter	Lux	Diameter	Lux	Diameter
White	5,360,000	0.1	1,340,000	0.19	595,536	0.29	335,000	0.38	148,889	0.58	53,600	0.96	13,400	1.92

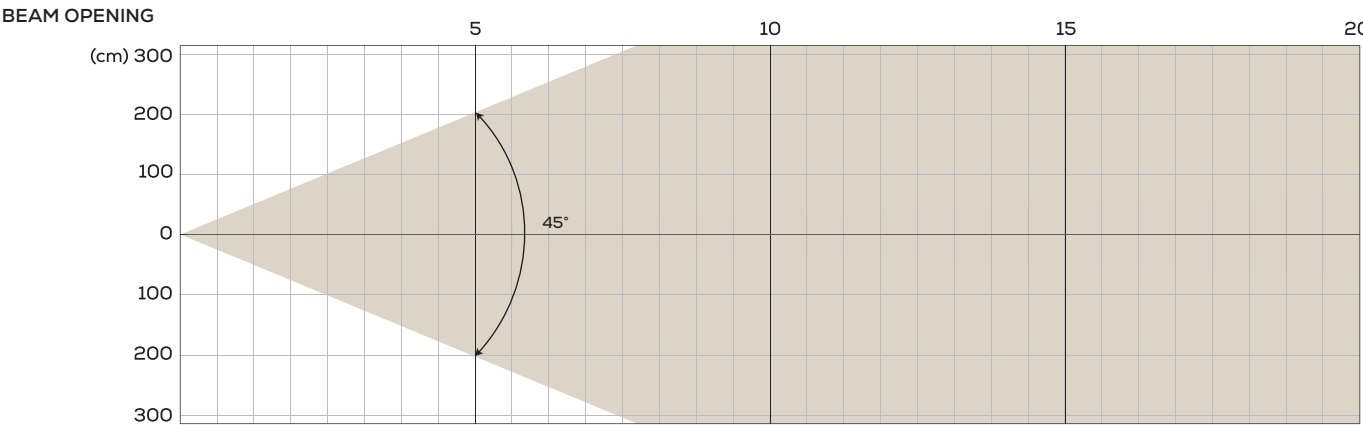
MIDDLE ZOOM



INTENSITY

Distance (m)	5		10		15		20	
unit	Lux	Diameter	Lux	Diameter	Lux	Diameter	Lux	Diameter
White	107,600	2.03	26,900	4.07	11,956	6.1	6,725	8.14

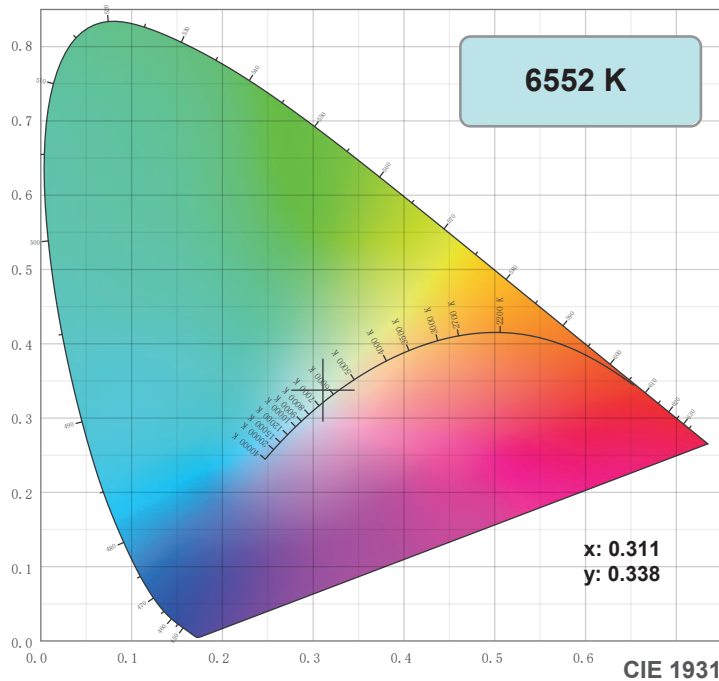
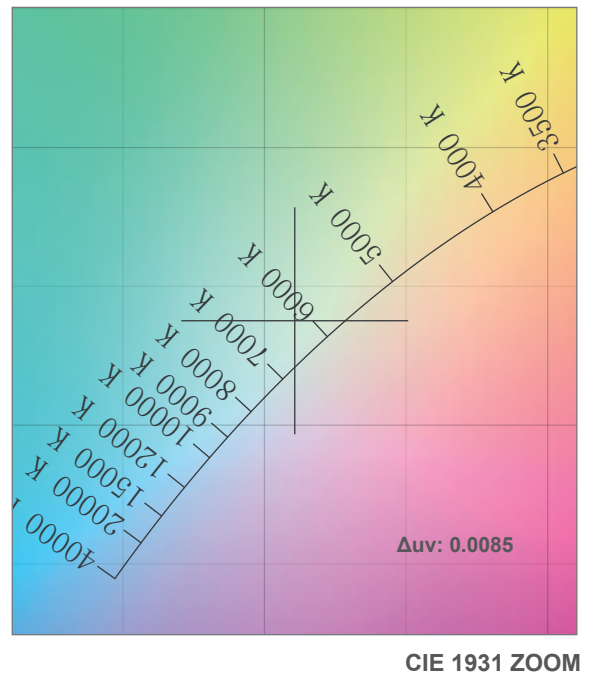
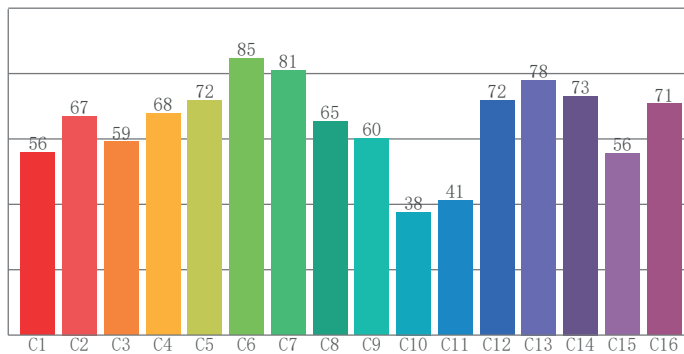
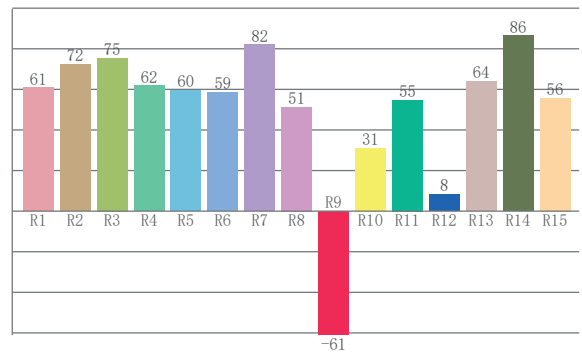
MAX ZOOM



INTENSITY

Distance (m)	5		10		15		20	
unit	Lux	Diameter	Lux	Diameter	Lux	Diameter	Lux	Diameter
White	25,964	4.14	6,491	8.28	2,885	12.43	1,623	16.57

COLOUR PARAMETERS

**TM-30: 64.4****CRI: 65.2 (R1-R8)**

CRI R values, only R1-R8 are used to calculate final CRI value

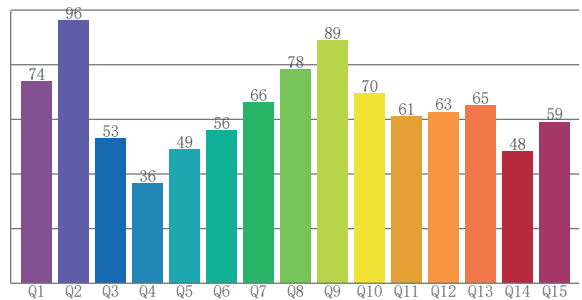
R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15
60.8	72.4	75.5	61.8	59.6	58.6	82.1	51.3	-61.1	31.1	54.8	8.3	63.9	86.5	55.7

TM30 C values, 16 binned values out of total of 99 C values

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16
55.8	67.0	59.1	67.9	71.9	84.7	81.0	65.4	60.2	37.6	41.1	71.8	78.0	73.1	55.6	70.8

CQS Q values

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15
73.8	96.2	53.0	36.5	49.1	56.0	66.2	78.2	89.0	69.6	61.0	62.7	65.0	48.3	59.1

CQS: 61.1

Color parameters

Color temperature	Color rendering index	Red component	Color fidelity	Color gamut	Color quality scale	Color coordinate cie 1931	Color coordinate cie 1931	Color coordinate	Color coordinate	Color deviation from black body
CCT	CRI	CRI R9	TM30 Rf	TM30 Rg	CQS	x	y	u	v	Δuv
6552 K	65.2	-61.1	64.4	84.3	61.1	0.311	0.338	0.193	0.315	0.0085

TM-30

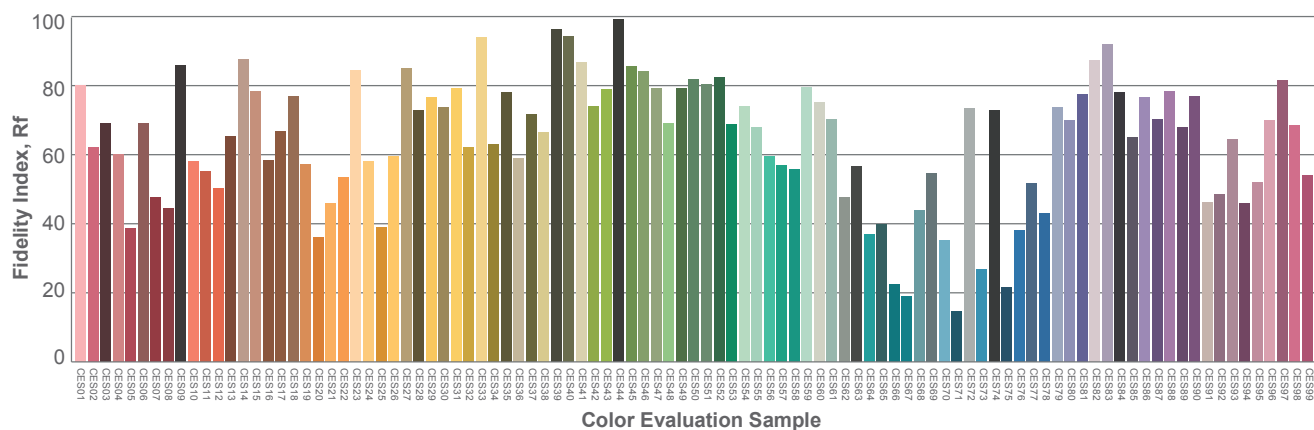
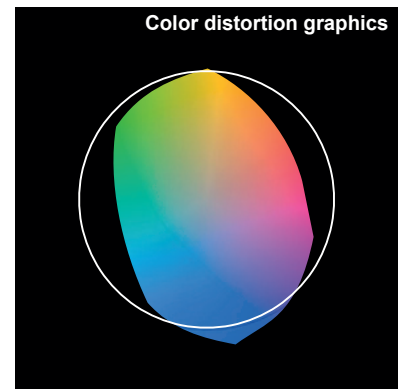
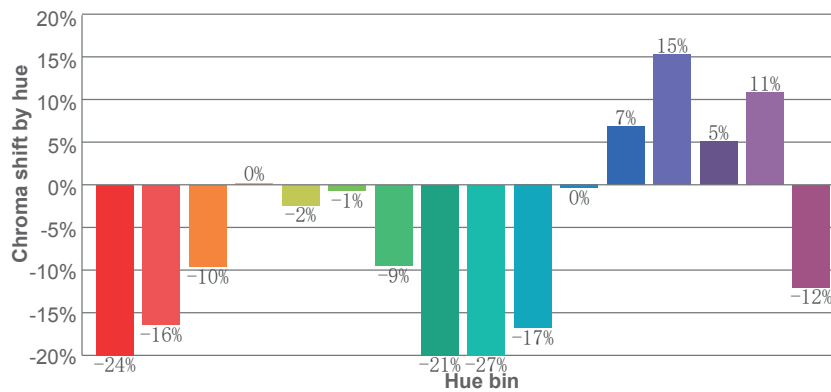
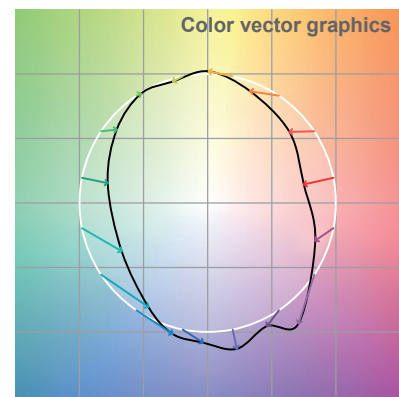
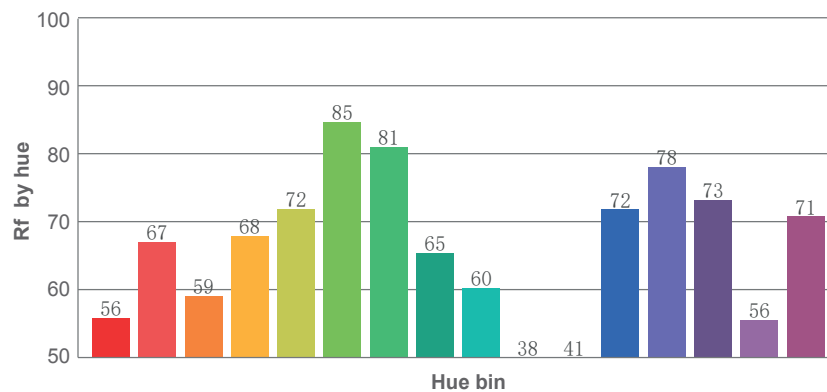
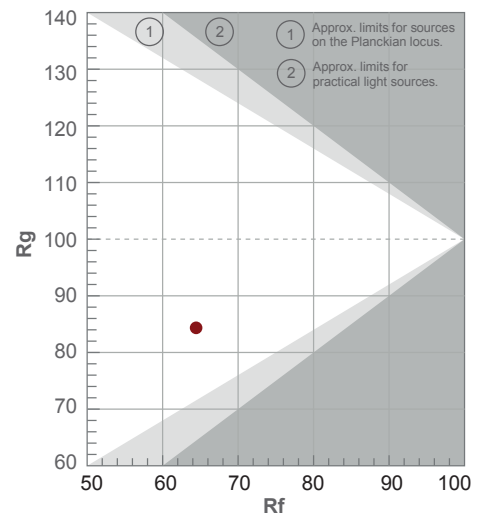
Rf 64.4

Fidelity index Rf

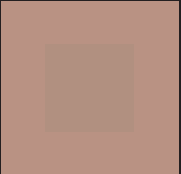
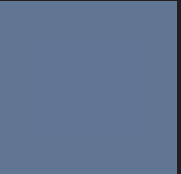
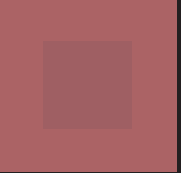
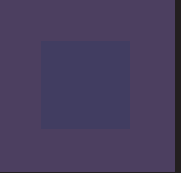
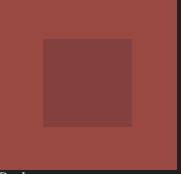
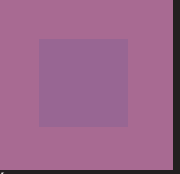
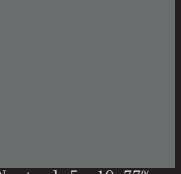
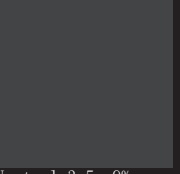
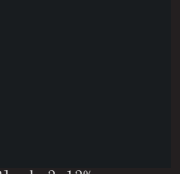
Rg 84.3

Gamut index Rg

Hue Bin	R_f	Shifts (%)	
		Chroma	Hue
1	56	-24%	-1%
2	67	-16%	10%
3	59	-10%	21%
4	68	0%	19%
5	72	-2%	8%
6	85	-1%	-4%
7	81	-9%	-8%
8	65	-21%	0%
9	60	-27%	24%
10	38	-17%	41%
11	41	0%	34%
12	72	7%	17%
13	78	15%	0%
14	73	5%	-15%
15	56	11%	-38%
16	71	-12%	-12%



TLCI-38

					
Dark skin	Light skin	Blue sky	Foliage	Blue flower	Bluish green
					
Orange	Purplish blue	Moderate red	Purple	Yellow green	Orange yellow
					
Blue	Green	Red	Yellow	Magenta	Cyan
					
White 90.01%	Neutral 8, 59.1%	Neutral 6.5, 36.2%	Neutral 5, 19.77%	Neutral 3.5, 9%	Black 3.13%

 DISCLAIMER

Errors and omissions for all information given in this specification sheet are possible. All information is subject to change without prior notice.